BT1010 Introduction to Life Sciences

Lecture 3: Biomolecules_Nucleic Acids, DNA-Based Information Technologies, Cloning

29/10/2025



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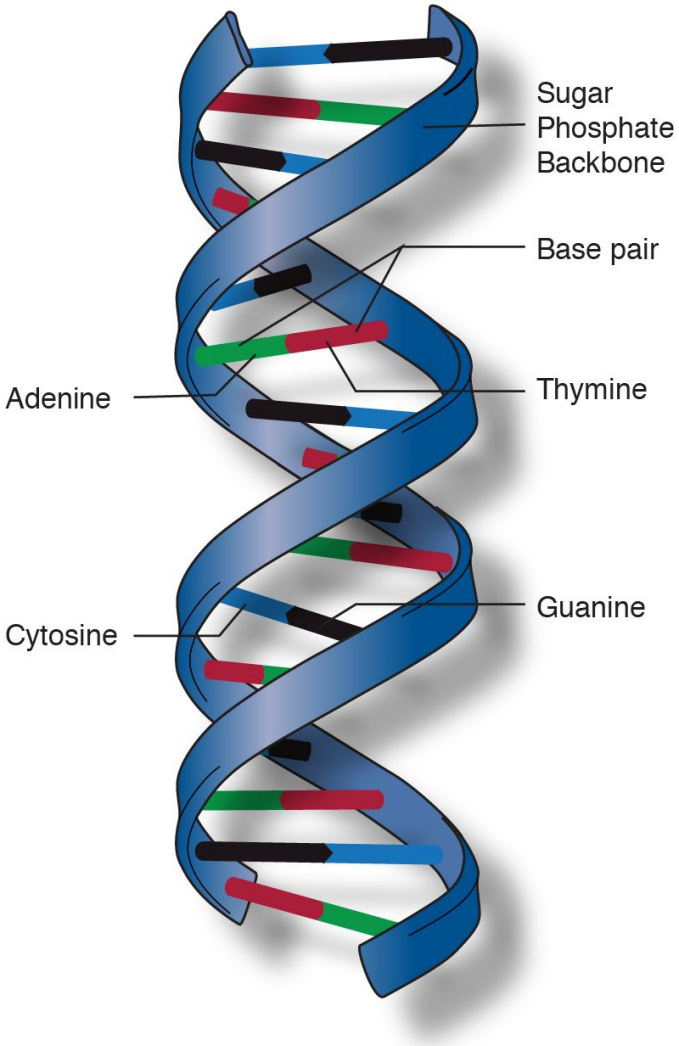
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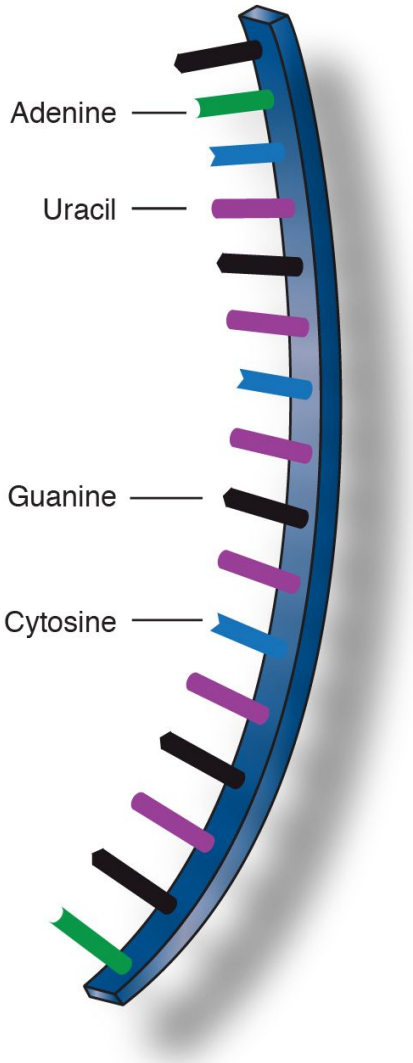
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Nucleic Acids



Deoxyribonucleic acid
(DNA)

Genetic material



Ribonucleic acid
(RNA)

See how DNA and proteins work together:
https://www.youtube.com/watch?v=7Hk9jct2ozY&t=67s

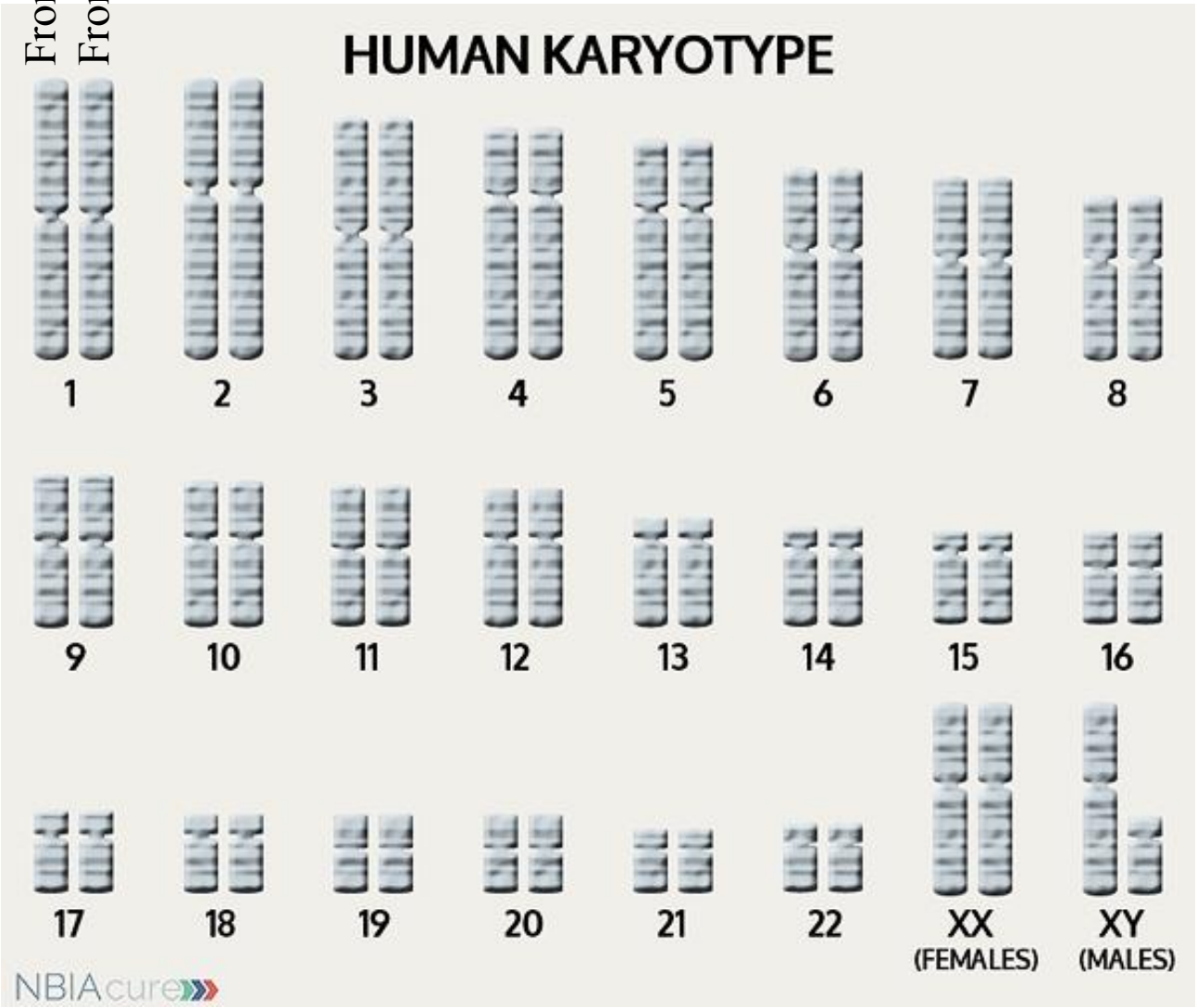
Why do we look similar to our parents or siblings?

Genetic Inheritance

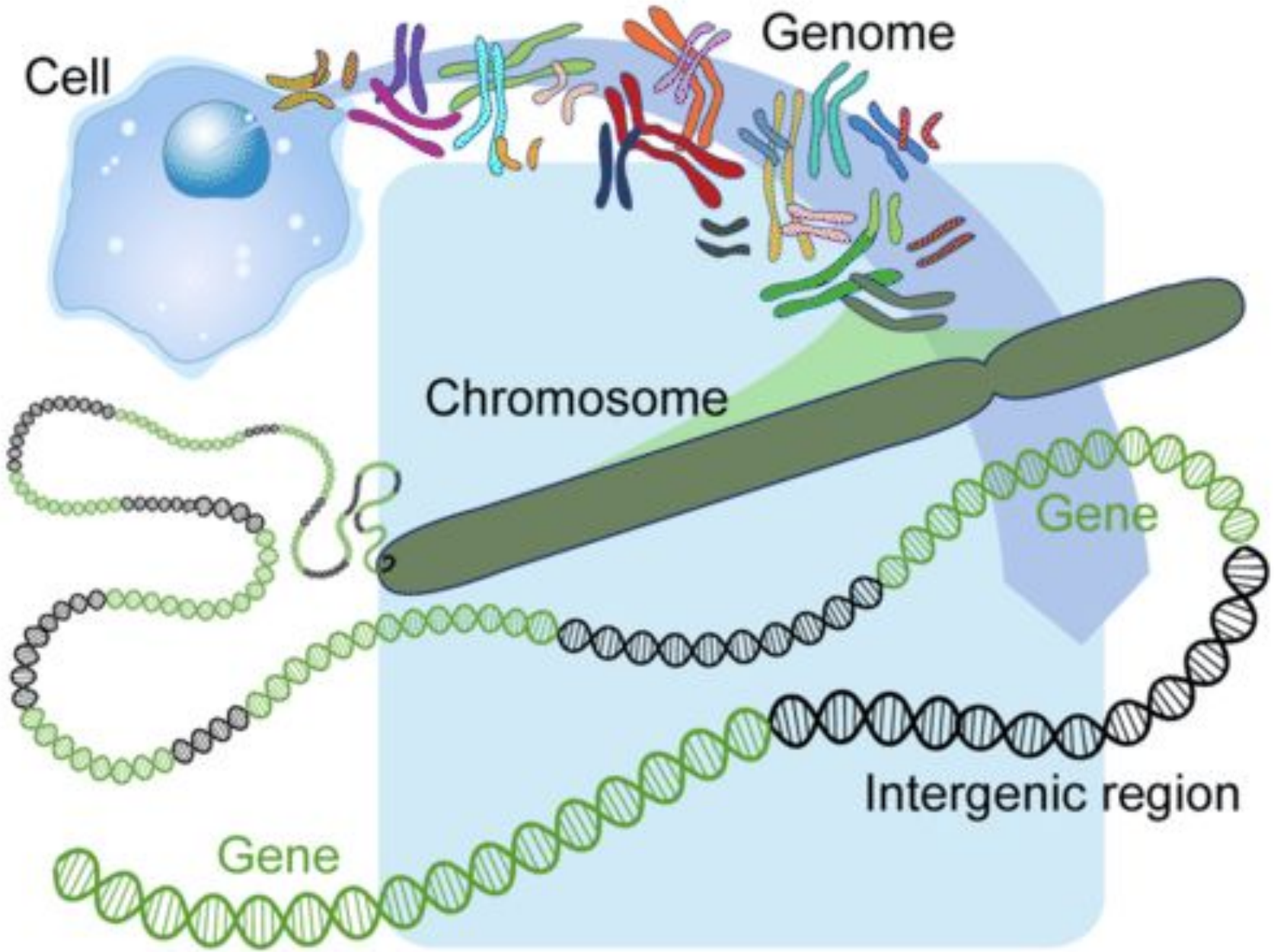


Human Chromosomes

From father
From mother



What is the difference between a DNA, Gene, Chromosome and Genome?



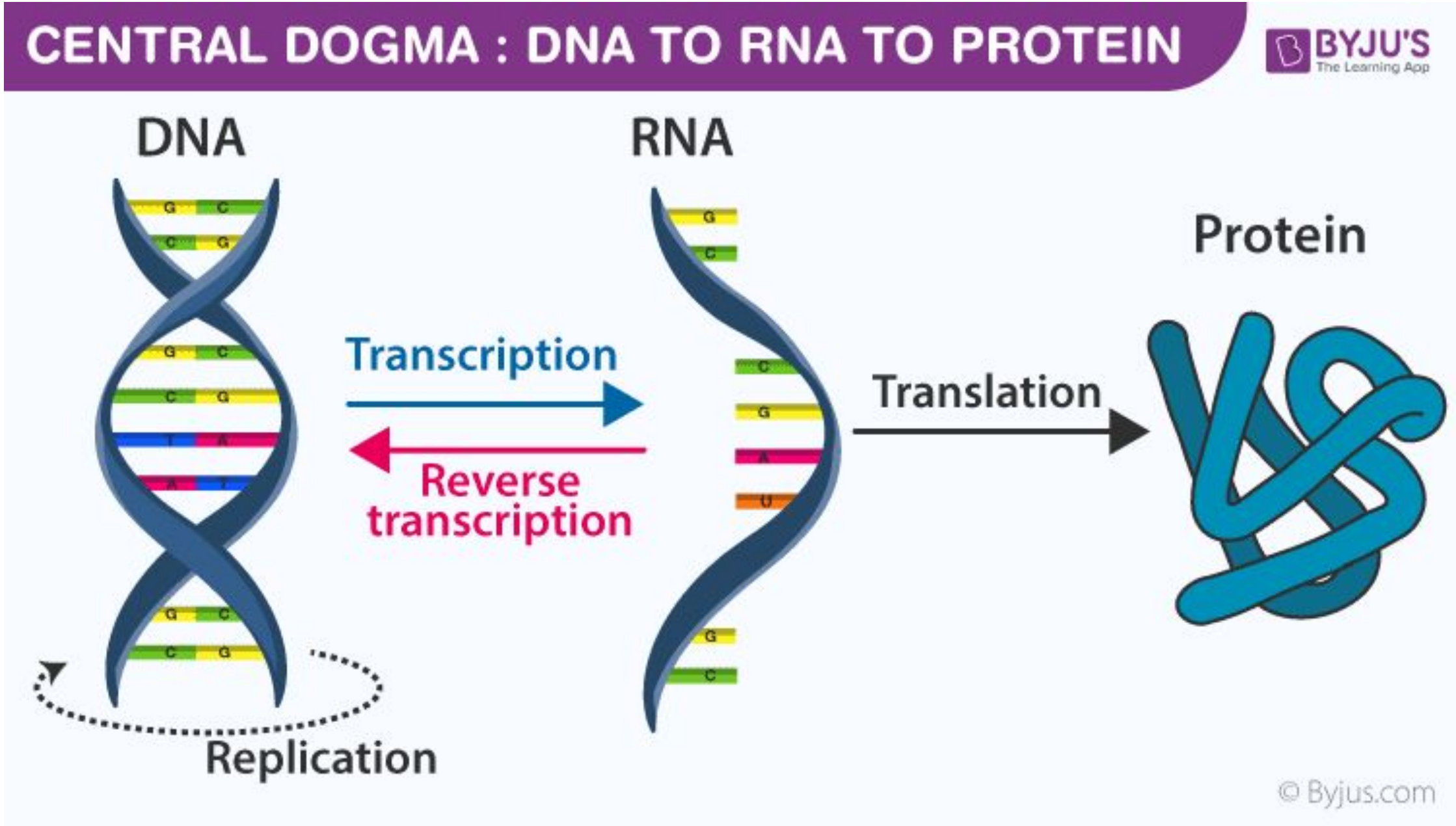
Introduction to Nucleic Acids

DNA: Deoxyribonucleic Acid

RNA: Ribonucleic Acid

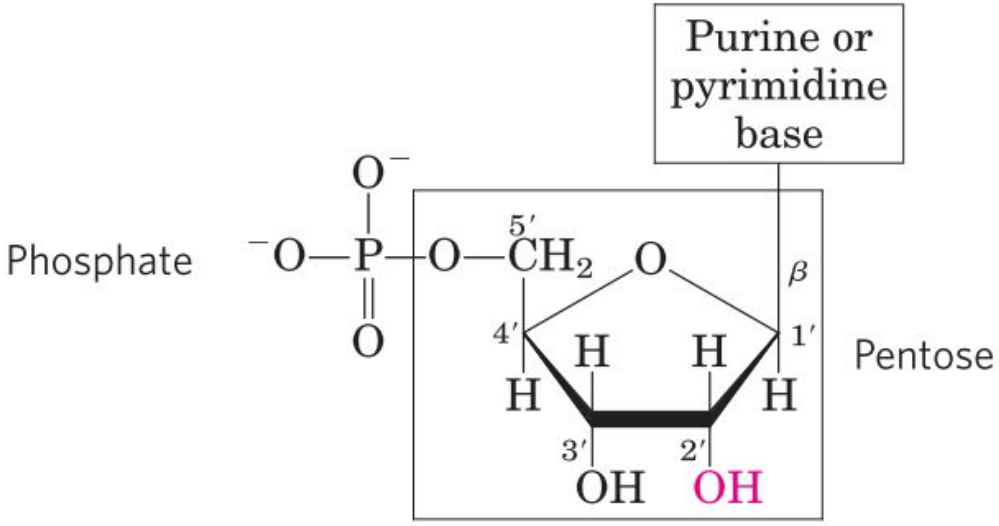
Nucleotides are the building blocks of nucleic acids.

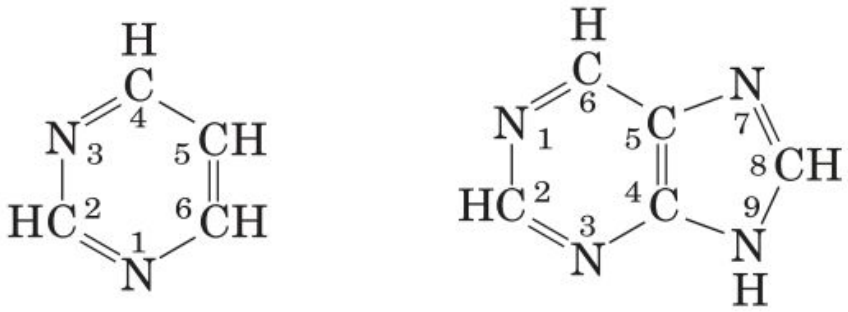
The amino acid sequences of proteins are specified by a nucleotide sequence in the cell's DNA.


CENTRAL DOGMA : DNA TO RNA TO PROTEIN
BYJU'S
The Learning App
DNA
RNA
Protein
Transcription
Reverse transcription
Translation
Replication
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- Nucleotides have three characteristic components: 1) a nitrogenous base, 2) a pentose, 3) one or more phosphates.

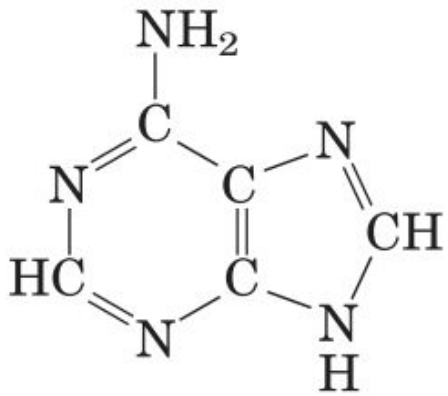
Structure of nucleotides



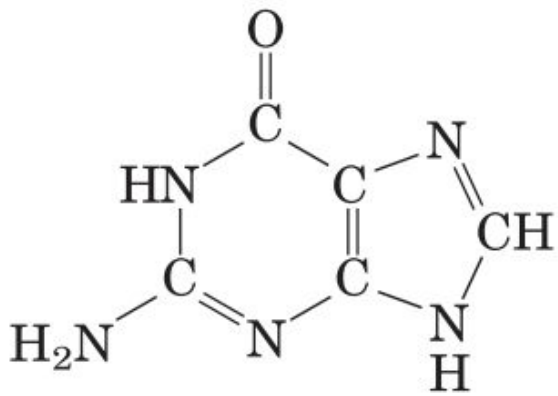


The nitrogenous bases are derivatives of two parent compounds, pyrimidine and purine.

Major purine and pyrimidine bases of nucleic acids

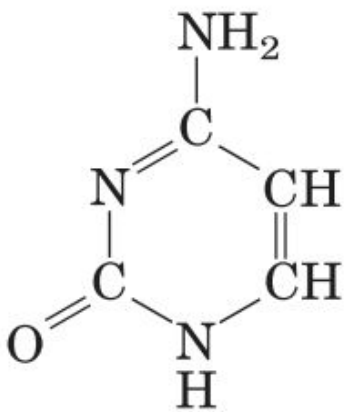


Adenine

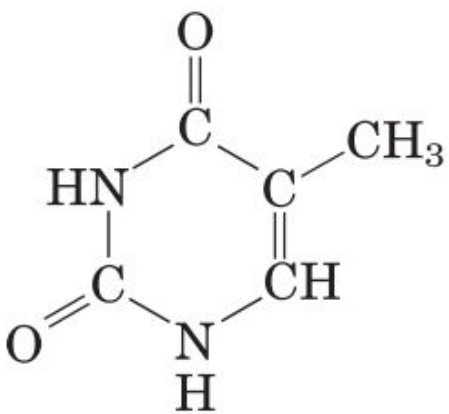


Guanine

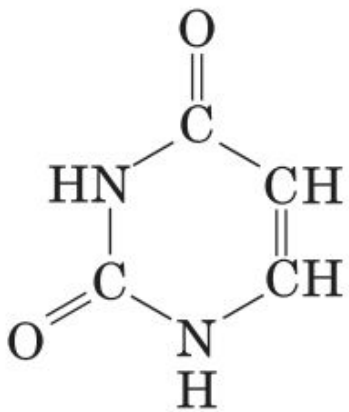
Purines



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines

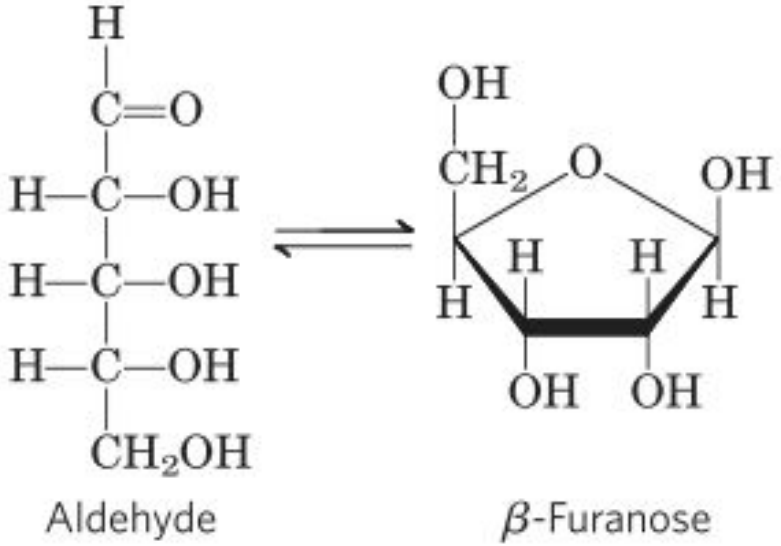
- A nucleotide without a phosphate group is called a nucleoside.

TABLE 8–1 Nucleotide and Nucleic Acid Nomenclature

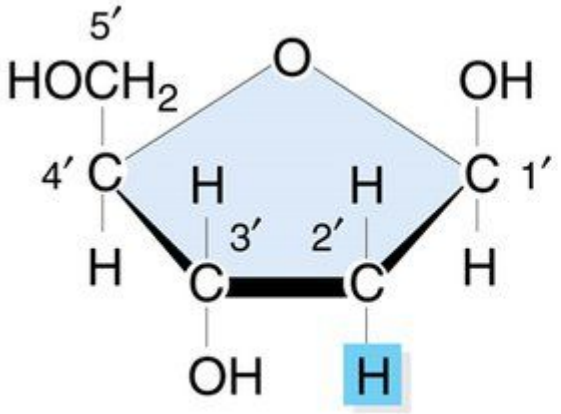
Base	Nucleoside	Nucleotide	Nucleic acid
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA

Types of pentose sugars

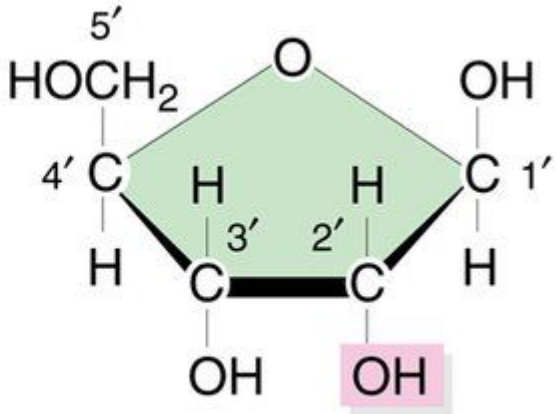
- Nucleic acids have two kinds of pentoses.
- DNA contains 2’-deoxy-D-ribose
- RNA contains D-ribose



Aldehyde β-Furanose

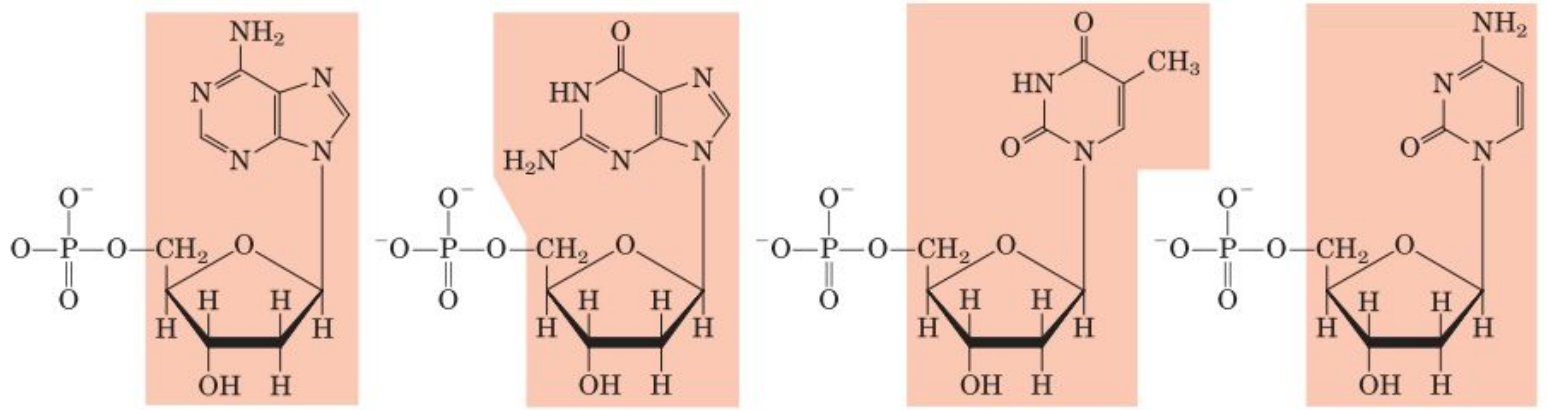


Deoxyribose



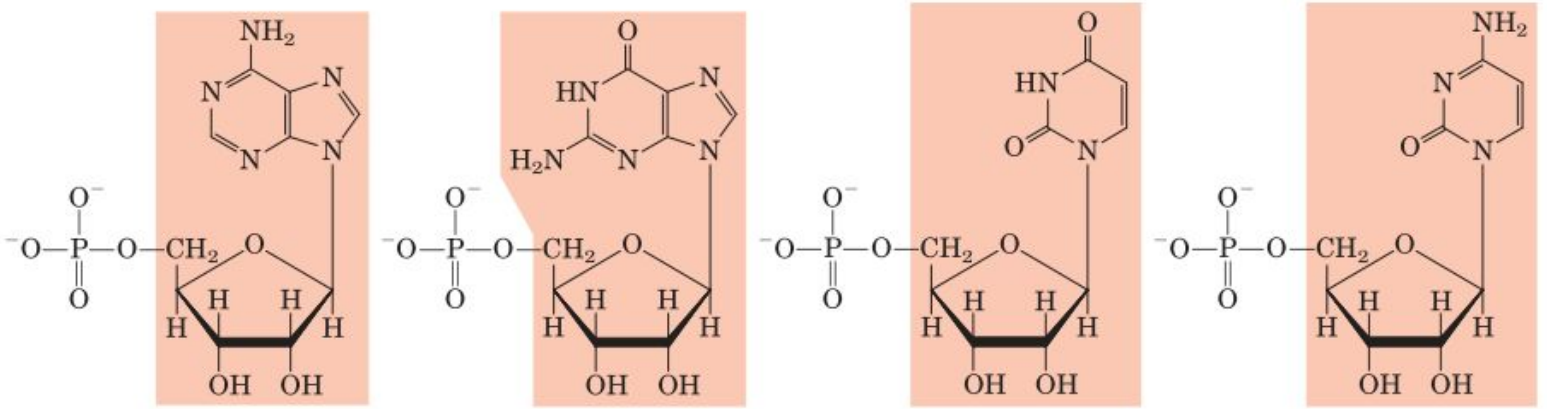
Ribose

Deoxyribonucleotides and ribonucleotides of nucleic acids



Nucleotide:	Deoxyadenylate (deoxyadenosine 5′-monophosphate)	Deoxyguanylate (deoxyguanosine 5′-monophosphate)	Deoxythymidylate (deoxythymidine 5′-monophosphate)	Deoxycytidylate (deoxycytidine 5′-monophosphate)
Symbols:	A, dA, dAMP	G, dG, dGMP	T, dT, dTMP	C, dC, dCMP
Nucleoside:	Deoxyadenosine	Deoxyguanosine	Deoxythymidine	Deoxycytidine

(a) Deoxyribonucleotides

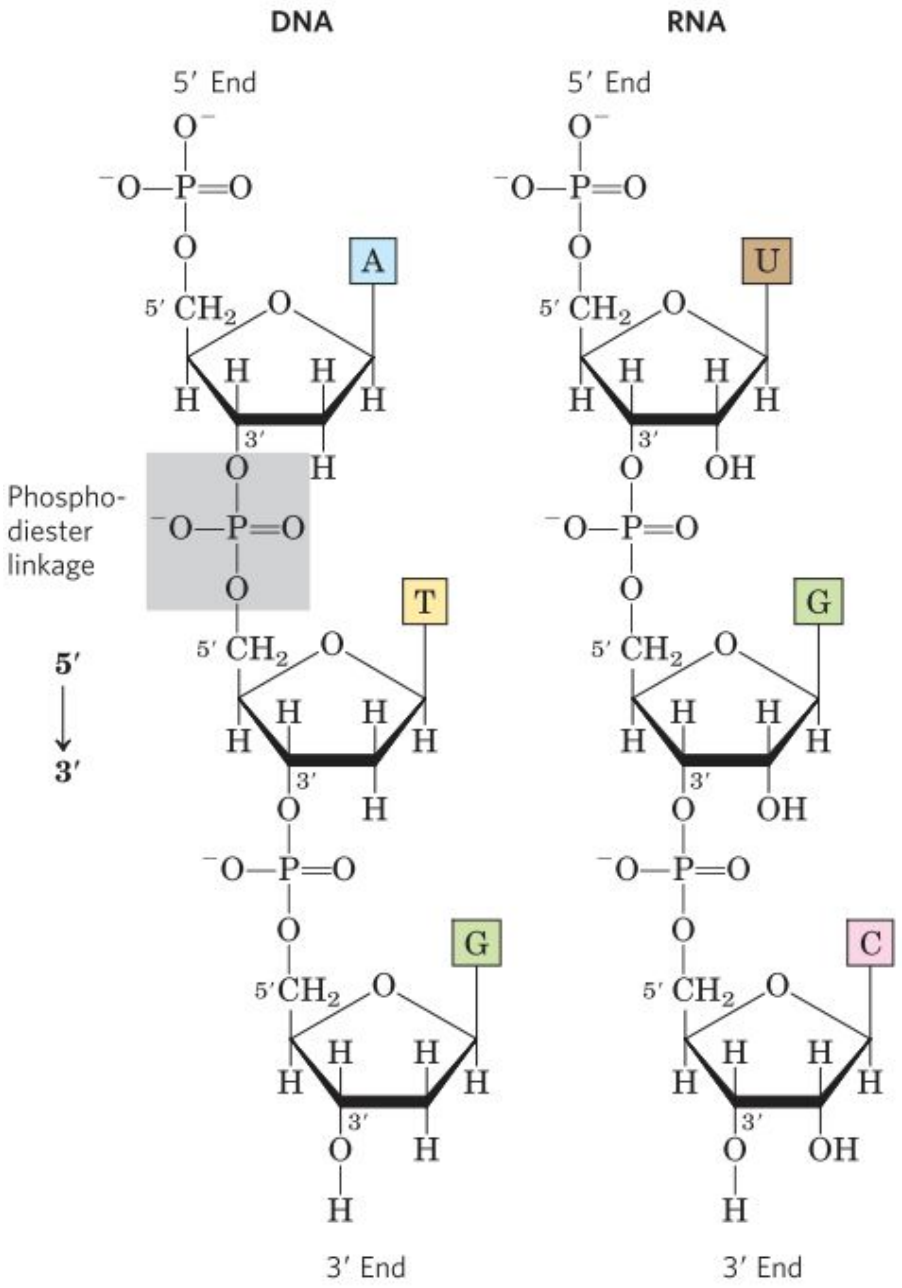


Nucleotide:	Adenylate (adenosine 5′-monophosphate)	Guanylate (guanosine 5′-monophosphate)	Uridylate (uridine 5′-monophosphate)	Cytidylate (cytidine 5′-monophosphate)
Symbols:	A, AMP	G, GMP	U, UMP	C, CMP
Nucleoside:	Adenosine	Guanosine	Uridine	Cytidine

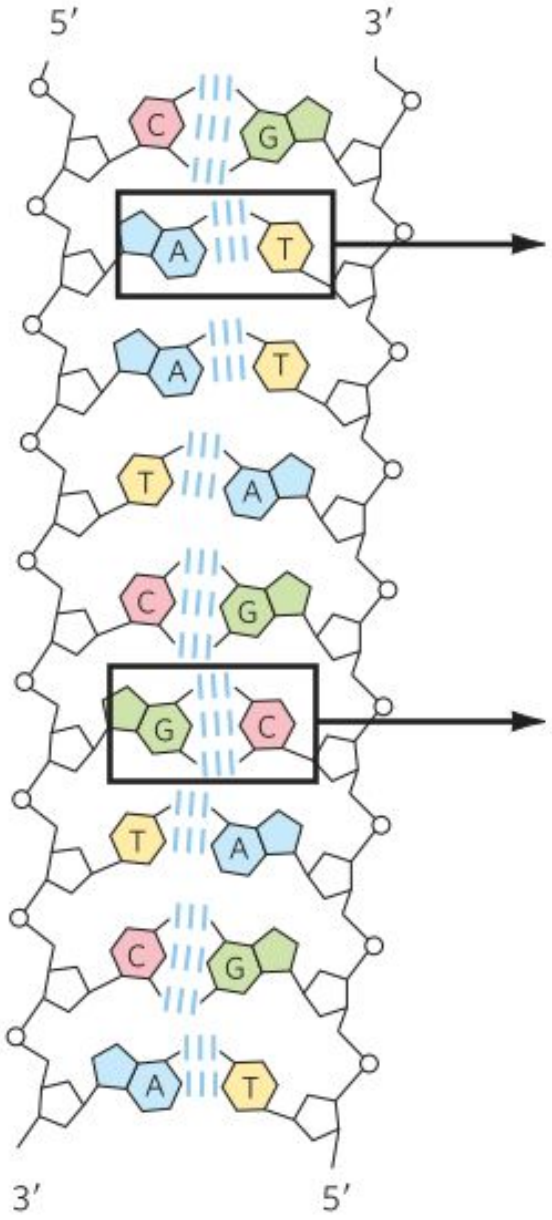
(b) Ribonucleotides

Phosphodiester bonds link successive nucleotides in nucleic acids

- The successive nucleotides of both DNA and RNA are covalently linked through phosphate-group “bridges”, in which the 5’-phosphate group of one nucleotide is joined to the 3’-hydroxyl group of the next nucleotide, creating a phosphodiester linkage.



Hydrogen-bonding patterns in the base pairs



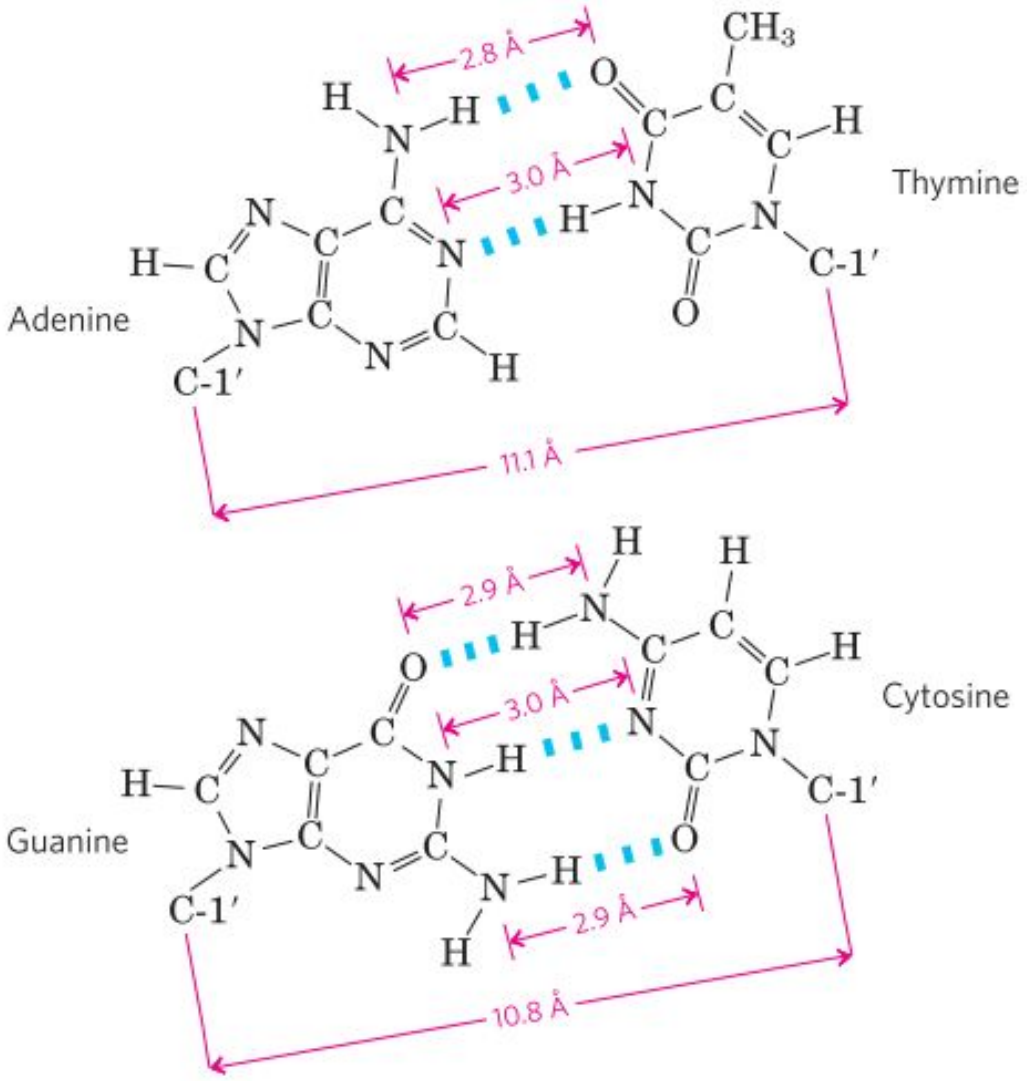


FIGURE 8–11 Hydrogen-bonding patterns in the base pairs defined by Watson and Crick. Here as elsewhere, hydrogen bonds are represented by three blue lines.

History and Discovery of DNA

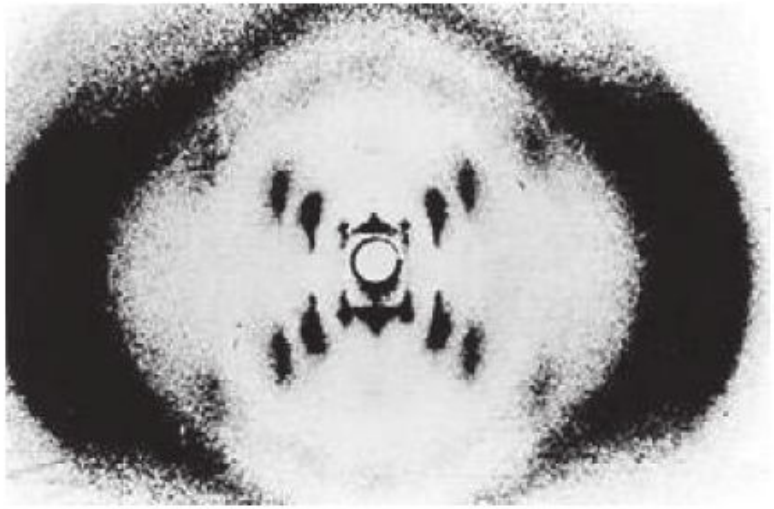
- DNA was first isolated and characterized by Friedrich Miescher in 1868. He called the phosphorous-containing substance "nuclein".
- In 1940, Avery, MacLeod and McCarty found that the DNA extracted from a virulent (disease-causing) strain of the bacterium Streptococcus pneumoniae and injected into a nonvirulent strain transformed the nonvirulent strain into a virulent strain.
- In 1952, Hershey and Chase studied the infection of bacterial cells by a virus (bacteriophage) with radioactively labeled DNA or protein confirmed that DNA, not protein, is the genetic material.
- In 1940, Erwin Chargaff provided following conclusions: (Chargaff's Rules)

1) The base composition of DNA generally varies from one species to another.
2) DNA specimens isolated from different tissues of the same species have the same base composition.
3) The base composition of DNA in a given species does not change with an organisms's age, nutritional status, or changing environment.
4) In all cellular DNA, A=T and G=C. (A+G = T+C)

History and Discovery of DNA

- In 1950s, Rosalin Franklin and Maurice Wilkins used X-Ray diffraction to analyze DNA fiber.
- In 1953, James Watson and Francis Crick relied on this accumulated information about DNA and postulated a 3D structure of DNA.
- In 1962, Watson, Crick and Wilkins were awarded the Nobel Prize in Physiology or Medicine.

X-Ray diffraction pattern of DNA fibers





Rosalind Franklin, 1920-1958



Maurice Wilkins, 1916-2004

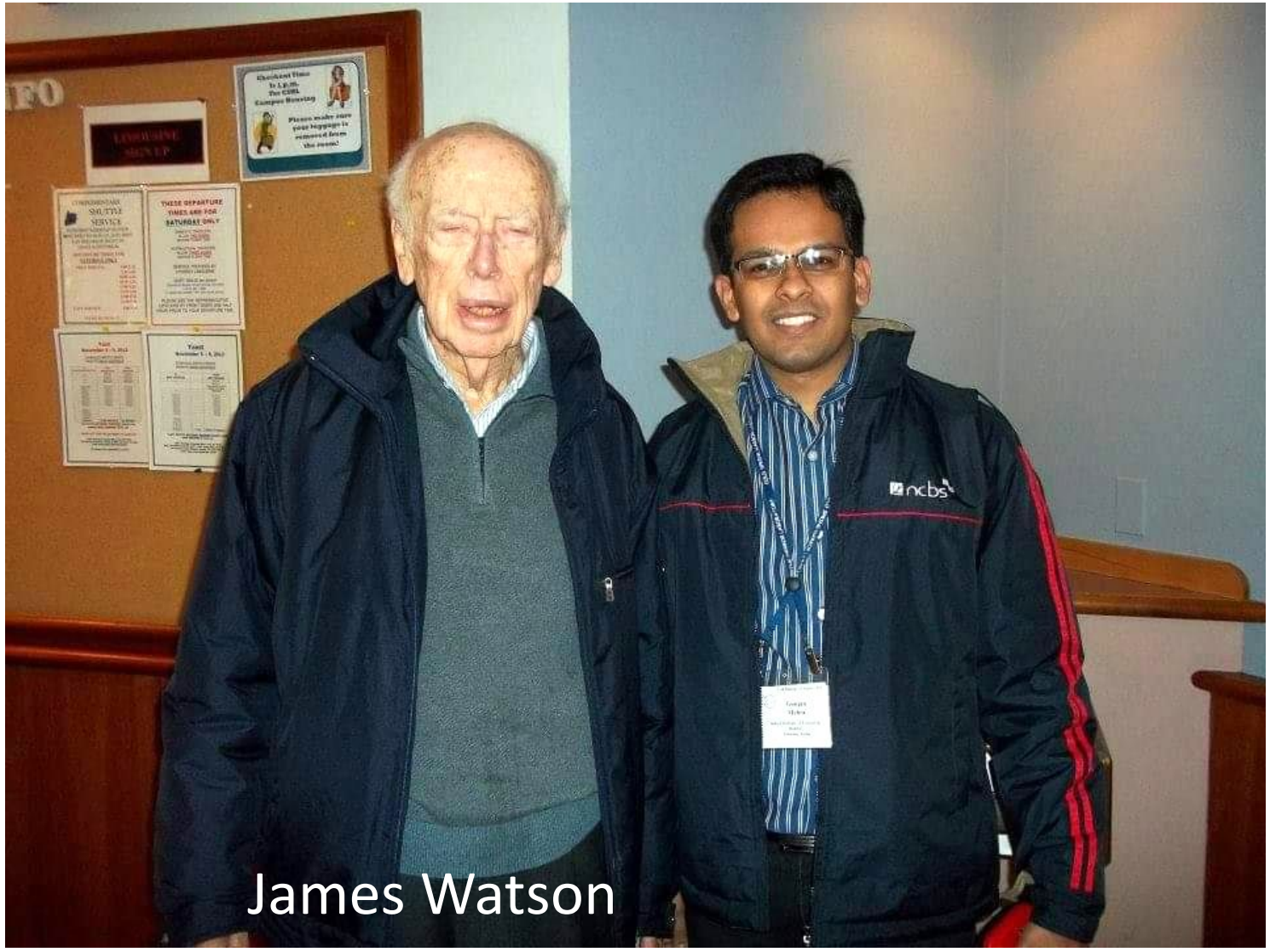


James D. Watson 1928-xxxx



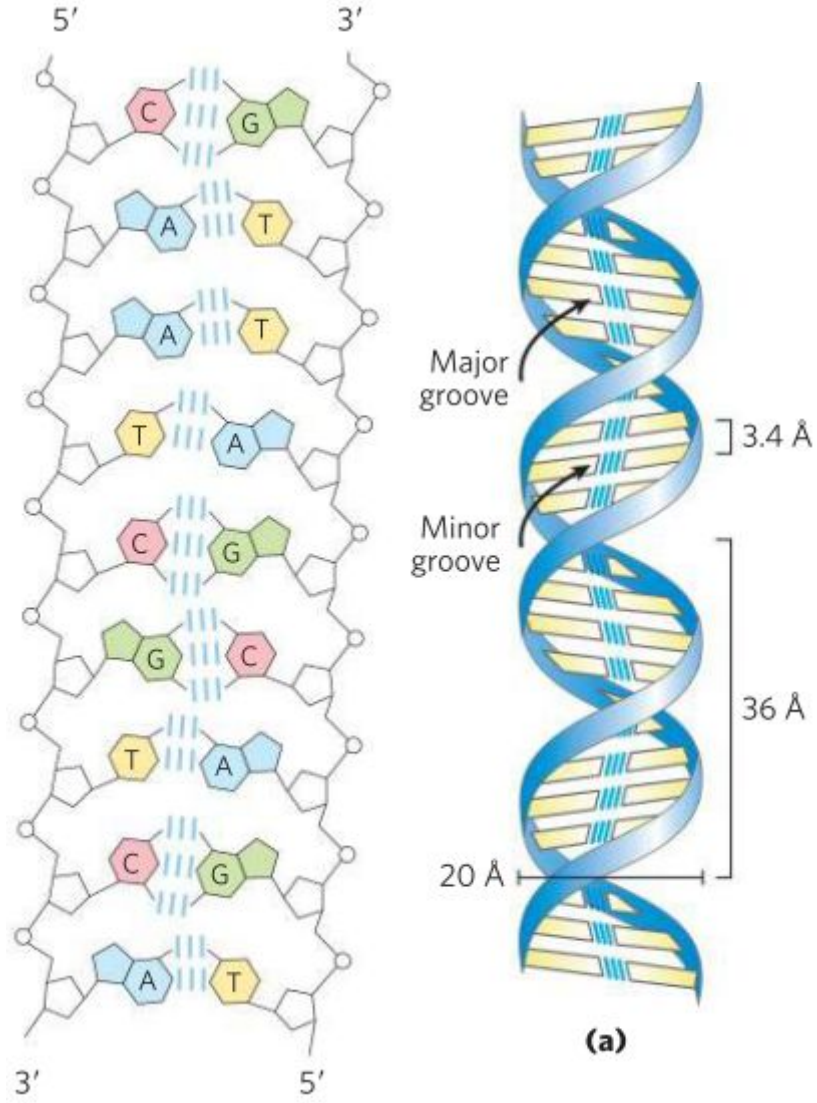
Francis Crick, 1916-2004

At Cold Spring Harbor Laboratory, New York, 2013



Structure of DNA

- It consists of two helical DNA chains wound around the same axis to form a right-handed double helix.
- The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water.
- The offset pairing of the two strands creates a major groove and minor groove on the surface of the duplex.
- Three hydrogen bonds form between G and C, whereas only two hydrogen bonds form between A and T.
- Both strands of DNA are in antiparallel orientation.
- Vertically stacked bases are 3.4 A apart.



RNA (Ribonucleic Acid)

- RNA is the second major form of nucleic acid in cells.
- RNA carries genetic information from DNA to the protein biosynthetic machinery of the ribosome.
- By the process of transcription, RNA is synthesized from the DNA template.
- Three types of RNAs involved in protein synthesis:

1) mRNA: Messenger RNA
2) tRNA: Transfer RNA
3) rRNA: Ribosomal RNA

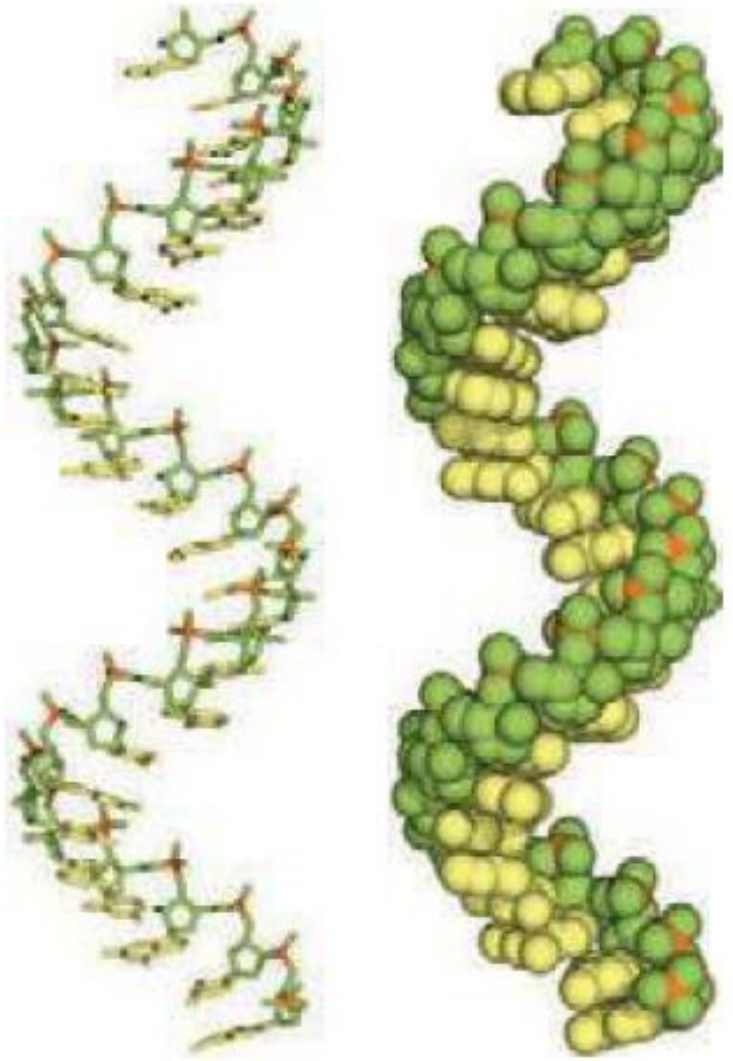
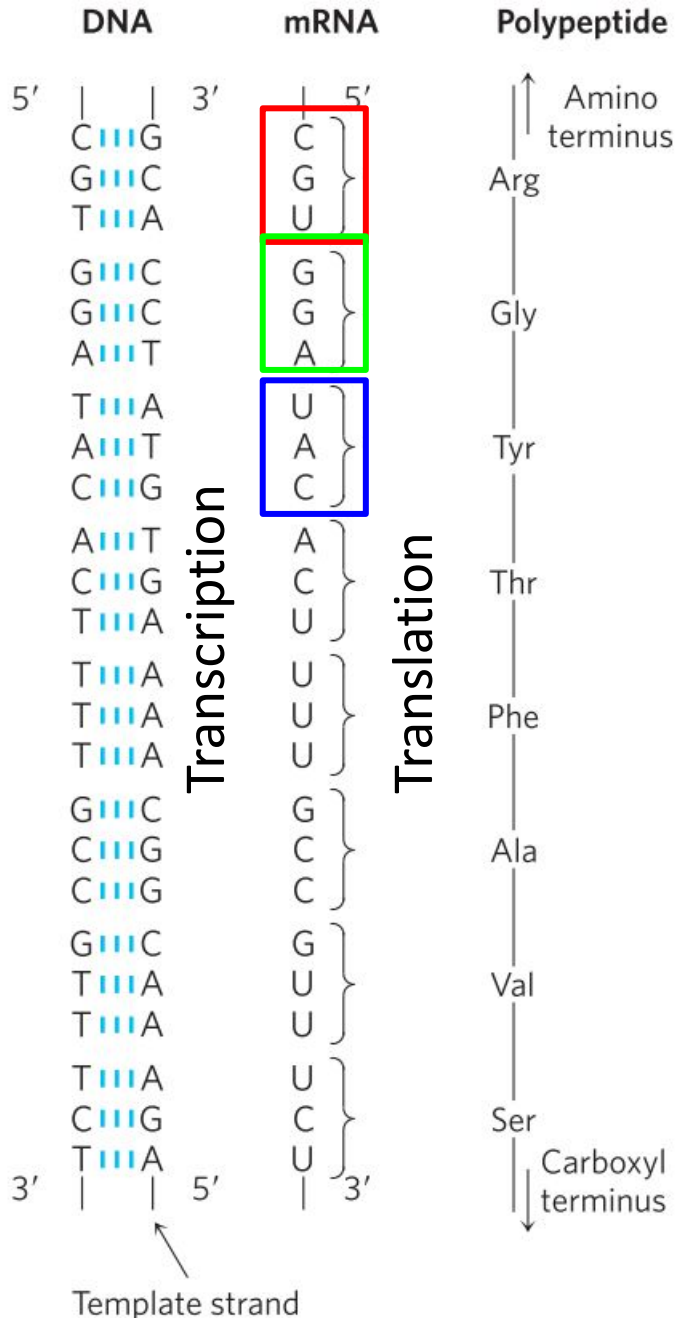


FIGURE 8–22 Typical right-handed stacking pattern of single-stranded RNA. The bases are shown in yellow, the phosphorus atoms in orange, and the riboses and phosphate oxygens in green. Green is used to represent RNA strands in succeeding chapters, just as blue is used for DNA.

How the polypeptide/protein chain is synthesized based on the DNA and RNA sequences?



Amino acid codon table

Second letter

First letter	U	C	A	G	Third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr **UAA Stop** **UAG Stop**	UGU Cys UGC Cys **UGA Stop** UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile **AUG Met**	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

DNA-based information technologies_Cloning

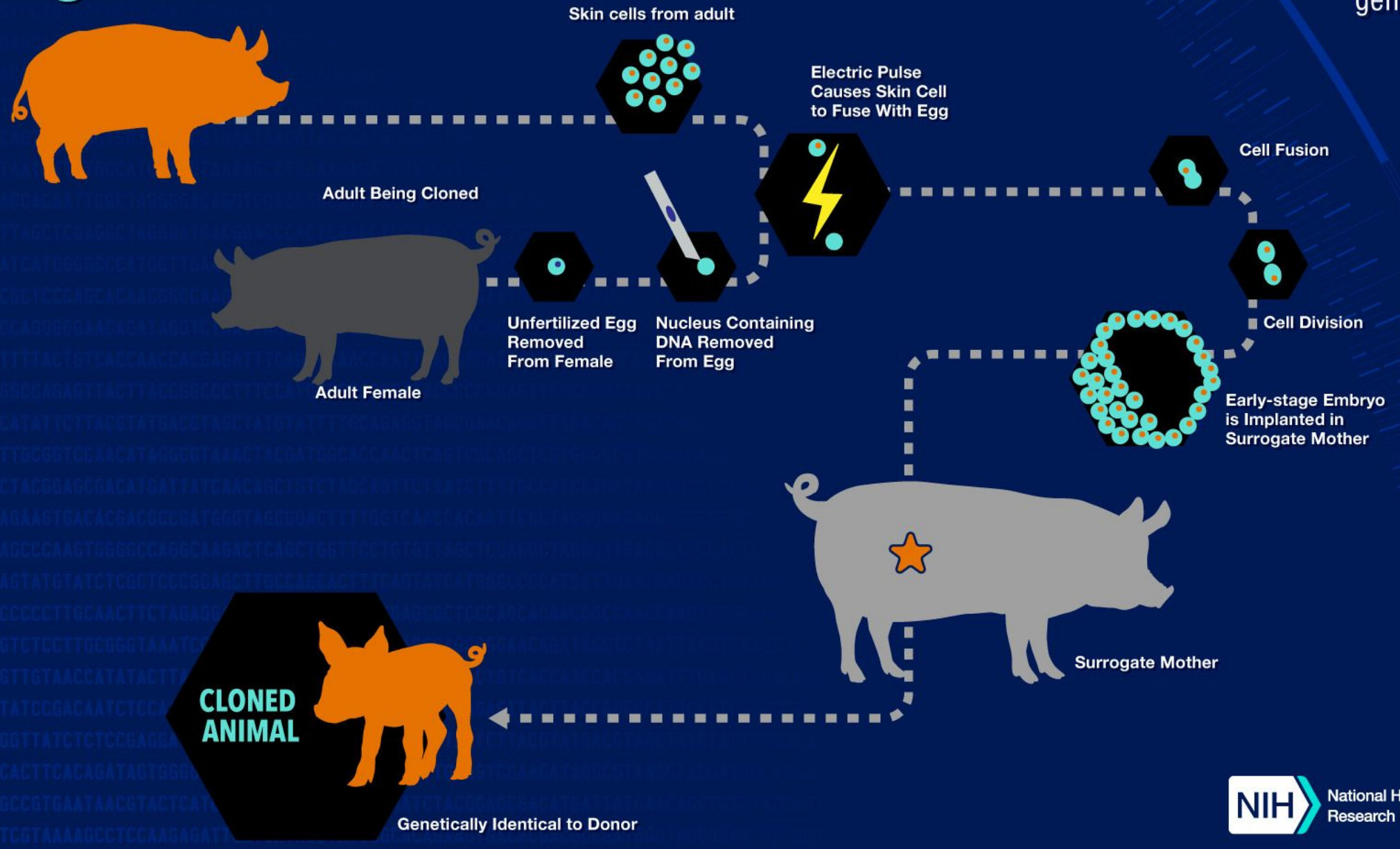
- Cloning is a technique scientists use to make exact genetic copies of living organisms.
- Genes, cells, tissues and even whole animal can be cloned.
- Dolly the sheep: was the first mammal to have been successfully cloned from an adult somatic cell in 1996 in Scotland.
- Dolly died at the age of 6 years, whereas the regular lifespan of sheep is ~12 to 13 years.
- Dolly was euthanized at the age of 6 years due to liver cancer and arthritis.



Dolly: 5 July 1996 – 14 February 2003
National Museum of Scotland

Cloning

NHGRI FACT SHEETS
genome.gov



NIH National Human Genome Research Institute

Ethical Issues of Cloning Animals

- There are a variety of ethical concerns regarding the possibilities of cloning, especially human cloning.
- While many views are religious in origin, there are also secular perspectives.
- Advocates support development of therapeutic cloning to generate tissues and whole organs to treat patients.
- Advocates for reproductive cloning believe that patients who cannot reproduce should have access to the technology.
- Opponents of cloning have concerns that the technology can be prone to abuse and how cloned individuals could integrate with families and with society at large.
- Religious groups believes that the technology can be misused as usurping “God’s place”. Also this technology uses embryos, so it destroys human life. Other religious groups support therapeutic cloning’s potential life-saving benefits.
- Cloning of animals is opposed by animal-groups due to the number of cloned animals that suffer from malformations before they die.

DNA Cloning / Molecular Cloning / Recombinant DNA Technology / Genetic Engineering

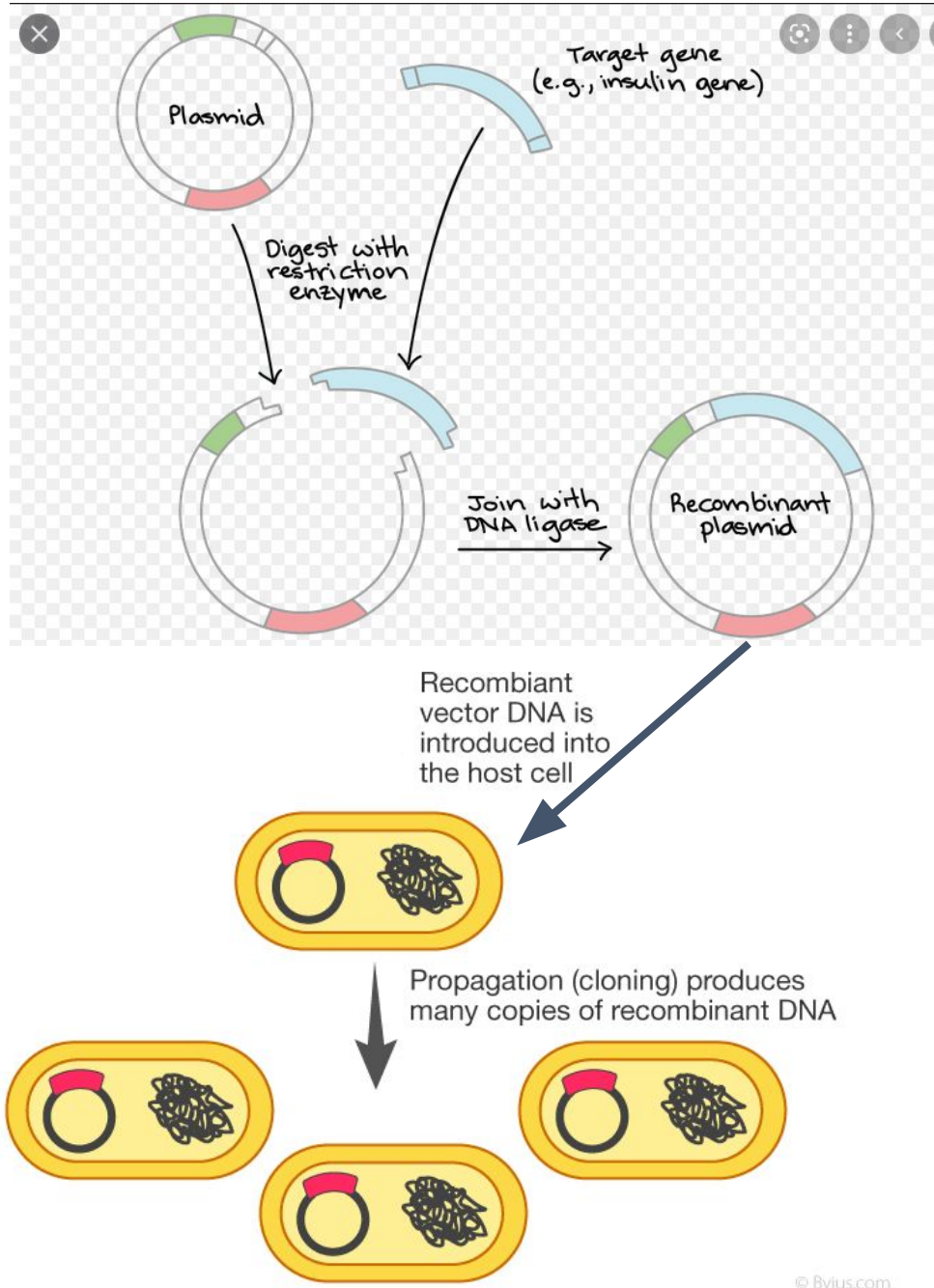
- DNA cloning is a molecular biology technique that makes many identical copies of a piece of DNA, such as a gene.

- In a typical DNA cloning procedure, the gene (perhaps a gene for a medically important human protein) is first inserted into a circular piece of DNA called a **plasmid**. The insertion is done using enzymes that "cut and paste" DNA, and it produces a molecule of **recombinant DNA.**

- Cloning of DNA entails five general procedures:

1) Cutting target DNA at precise locations. (Sequence-specific endonucleases) provide the necessary molecular scissors.
2) Selecting a small carrier DNA capable of self-replication. These DNAs are called cloning vectors or plasmids.
3) Joining two DNA fragments covalently. DNA ligase is an enzyme that links cloning vector and the DNA to be cloned. The resultant DNA molecule is known as recombinant DNA.
4) Moving recombinant DNA from the test tube to a host cell that will provide enzymatic machinery for DNA replication.
5) Selecting host cells that contain recombinant DNA.

- The methods used to accomplish these and related tasks are collectively refe to as recombinant DNA technology or Genetic engineering.



Cloning vectors (plasmids) allow amplification of inserted DNA segments

- Plasmids: A plasmid is a circular DNA molecule that replicates separately from the host chromosome.
- Their size ranges from 5000 to 400000 bp.
- To survive in the host cell, plasmids incorporate several specialized sequences that enable them to make use of the cell's resources for their own replication and gene expression.
- pBR322, constructed in 1977, is a good example of a plasmid with features useful in almost all cloning vectors.
  1) Origin of replication (ori), a sequence where replication is initiated by cellular enzymes. An associated regulatory system is present that limits replication to maintain pBR322 at a level of 10 to 20 copies per cell. (Two different plasmids cannot function in the same cell if they use the same ori.)
  1) Antibiotic resistance genes Tet^R (Tetracycline Resistance) and Amp^R (Ampicillin Resistance) allow selection of cells that contain the intact plasmid or a recombinant version of the plasmid.
  1) RE sites to be cut and insert foreign DNA.
  2) Small size of the plasmid (4361 bp) facilitates its entry into cells and the biochemical manipulation of the DNA.

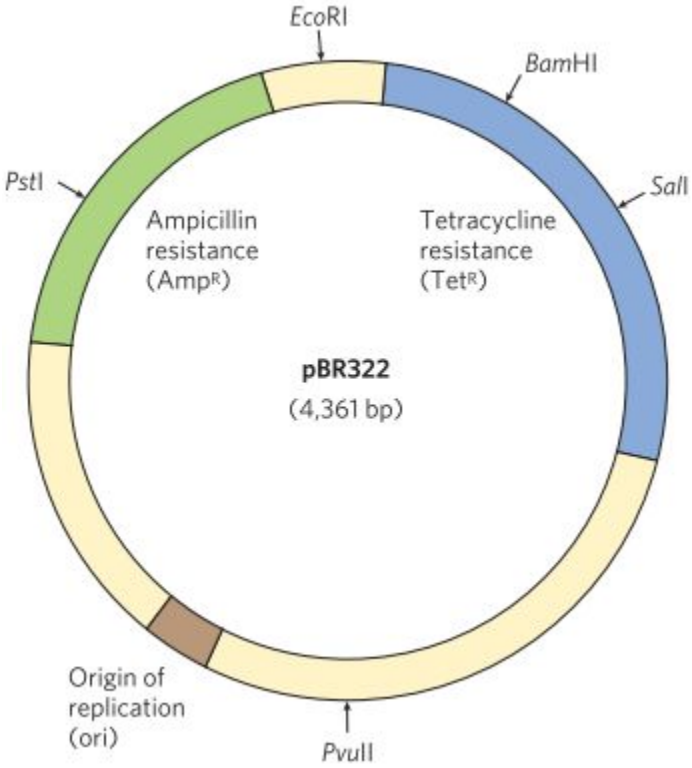


TABLE 9–1 Some Enzymes Used in Recombinant DNA Technology

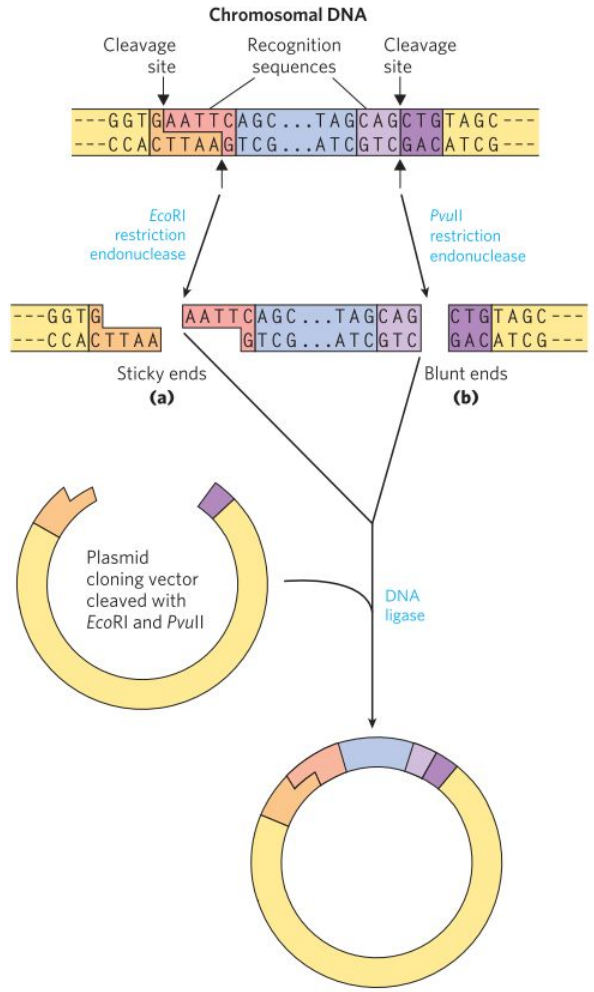
Enzyme(s)	Function
Type II restriction endonucleases	Cleave DNAs at specific base sequences
DNA ligase	Joins two DNA molecules or fragments
DNA polymerase I (*E. coli*)	Fills gaps in duplexes by stepwise addition of nucleotides to 3′ ends
Reverse transcriptase	Makes a DNA copy of an RNA molecule
Polynucleotide kinase	Adds a phosphate to the 5′-OH end of a polynucleotide to label it or permit ligation
Terminal transferase	Adds homopolymer tails to the 3′-OH ends of a linear duplex
Exonuclease III	Removes nucleotide residues from the 3′ ends of a DNA strand
Bacteriophage λ exonuclease	Removes nucleotides from the 5′ ends of a duplex to expose single-stranded 3′ ends
Alkaline phosphatase	Removes terminal phosphates from either the 5′ or 3′ end (or both)

DNA ligase joins the DNA molecule

- DNA ligase is used to join two DNA molecules. It forms phosphodiester bonds between the DNA molecules.

- A vector (plasmid) digested by *Eco*RI can be ligated to an insert digested by the same RE. It can not be ligated to an insert digested by *Bam*HI.

- Blunt ends can also be ligated, albeit less efficiently.



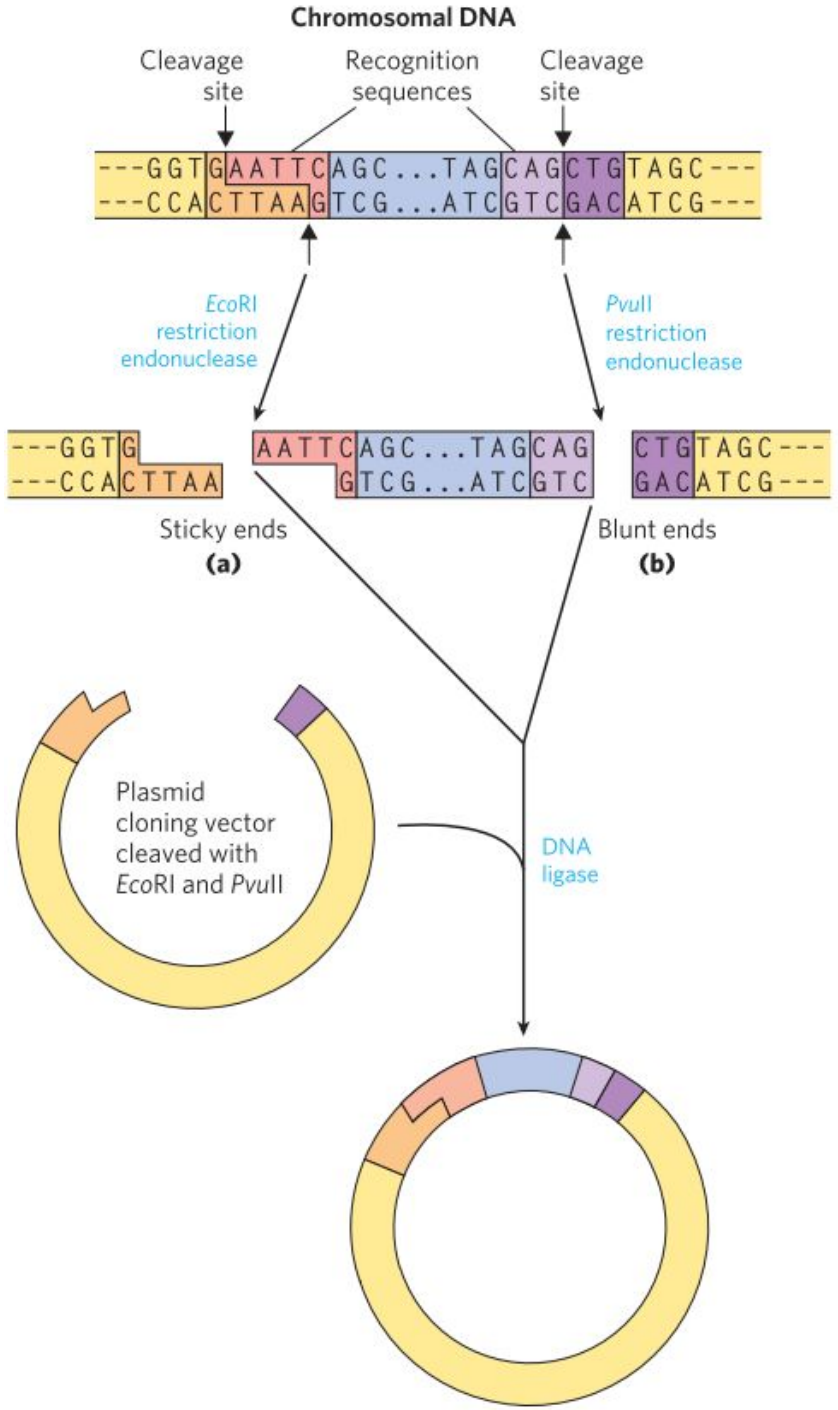


TABLE 9–2 Recognition Sequences for Some Type II Restriction Endonucleases

*Bam*HI	(5′) G G A T C C (3′) C C T A G G	*Hin*dIII	(5′) A A G C T T (3′) T T C G A A
*Cla*I	(5′) A T C G A T (3′) T A G C T A	*Not*I	(5′) G C G G C C G C (3′) C G C C G G C G
*Eco*RI	(5′) G A A T T C (3′) C T T A A G	*Pst*I	(5′) C T G C A G (3′) G A C G T C
*Eco*RV	(5′) G A T A T C (3′) C T A T A G	*Pvu*II	(5′) C A G C T G (3′) G T C G A C
*Hae*III	(5′) G G C C (3′) C C G G	*Tth*111I	(5′) G A C N N N G T C (3′) C T G N N N C A G

In general, a 6 bp sequence recognized by a RE (such as BamHI) would occur on average once every 4^6 (4096) bp.

An enzyme that recognize a 4 bp sequence would occur on an average once every 4^4 (256) bp.

How to introduce plasmids into bacterial cells?

There are two ways:

1) Transformation:
   The bacterial cells (often *E. coli*) and plasmid DNA are incubated together at 0 ^{o}C in a calcium chloride solution, then subjected to heat shock by rapid shifting the temperature to between 37 ^{o}C and 43 ^{o}C.

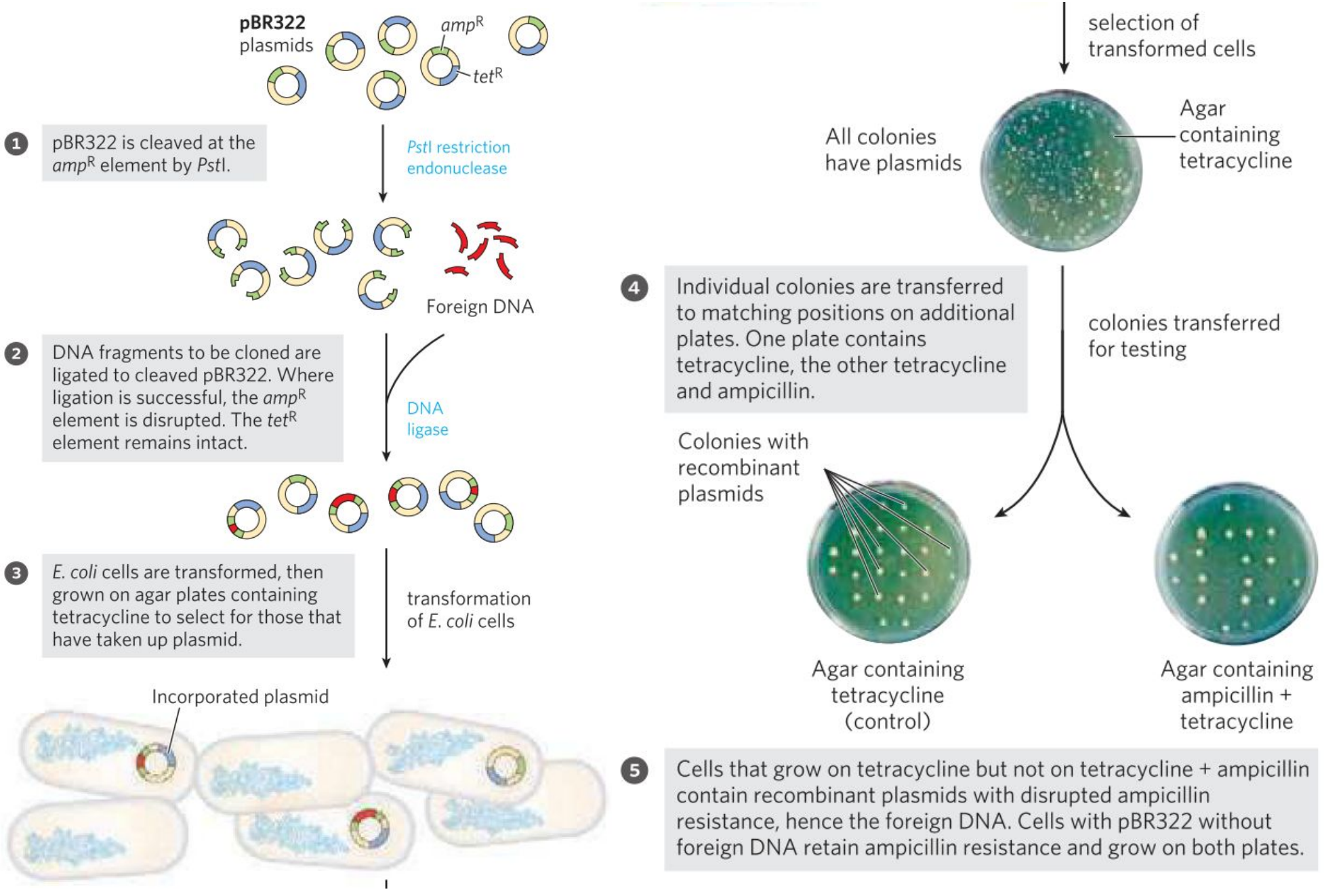
1) Electroporation:
   The bacterial cells (often *E. coli*) are incubated with plasmid DNA are subjected to high-voltage electric pulse. This approach transiently renders the bacterial membrane permeable to large molecules.

Regardless of the approach, relatively few cells take up the plasmid DNA, so a method is needed to identify those cells which contain the plasmid.

Plasmids usually contains one or two selectable markers (such as Tet^{R}, Amp^{R}).

Use of pBR322 to clone foreign DNA in *E. coli* and identify cells containing it.

A cloning strategy with pBR322

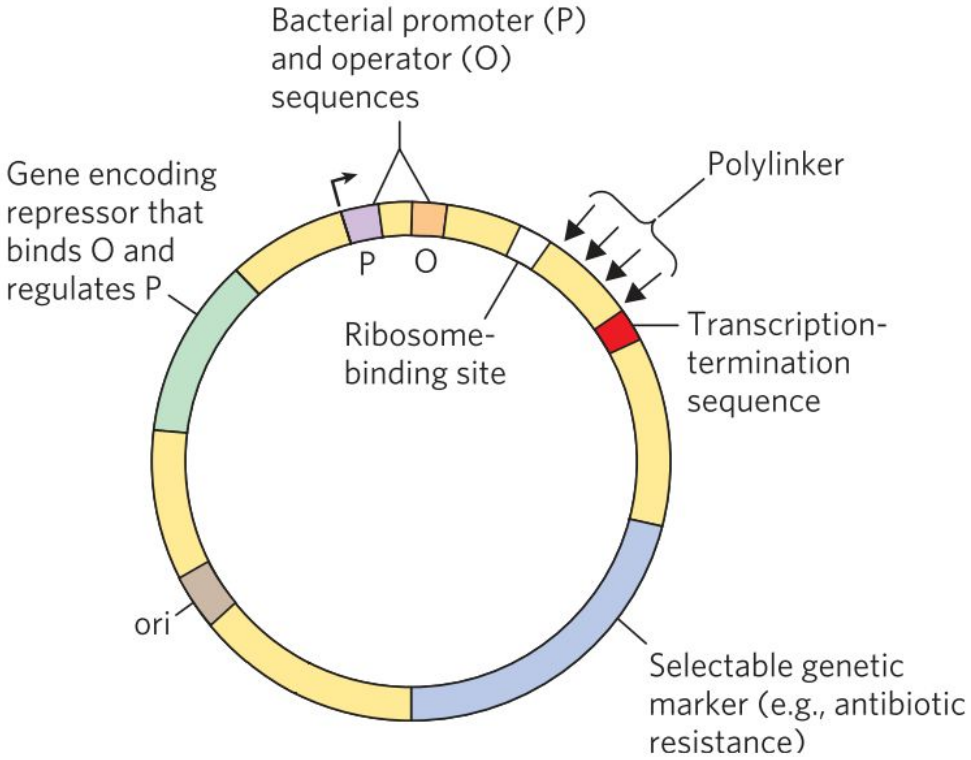


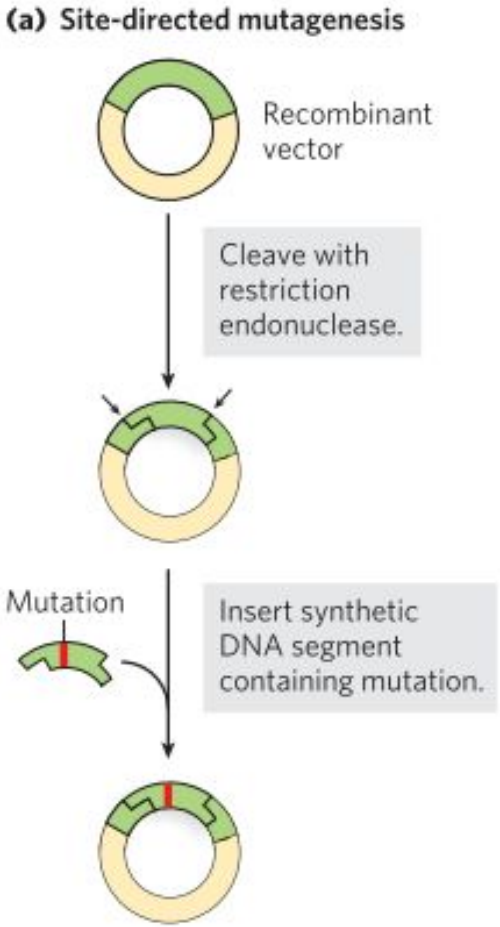
Cloned genes can be expressed to amplify protein production

- Frequently, the product of the cloned gene, rather than the gene itself, is of primary interest-particularly when the protein has commercial, therapeutic or research value.

- Cloning vectors with the transcription and translation signals needed for the regulated expression of a cloned gene are called **expression vectors**.

- Site-directed mutagenesis can be used to produce a mutant version of the protein.





Further reading

Book: Lehninger: Principles of Biochemistry, 6th edition. ISBN: 1464109621.

Chapter 8: Nucleotides and Nucleic Acids, Page 281-312.

Chapter 9: DNA-Based Information Technologies, Page 313-355.

Next class on 30/10/2025 (Thursday)

Restriction Endonucleases

- Restriction endonucleases (RE), also called restriction enzymes, recognize and cleave DNA at specific sequences (recognition sequences or restriction sites) to generate a set of smaller fragments.

- REs are found in a wide range of bacterial species. However, in its host cells, REs can't digest its DNA because of methylation of the DNA bases. This mechanism is known as Restriction Modification System.

- Three types of REs: Type I, Type II and Type III

- Type I and III are generally large, multisubunit complexes containing both the endonuclease and methylase activities. They require ATP for energy.

- Type I REs cleave DNA at random sites that can be more than 1000 bp from the recognition sequences.

- Type III REs cleave the DNA about 25 bp from the recognition sequences.

- Type II REs cleave the DNA within the recognition sequence itself. They don't need ATP. The recognition sequences are usually 4 to 6 bp long and they are palindromic.

Some of the Type II REs make staggered cuts on the two DNA strands, leaving two to four nucleotides of one strand unpaired at each resulting end. These unpaired strands are referred to as sticky ends.

DNA can occur in different three-dimentional forms

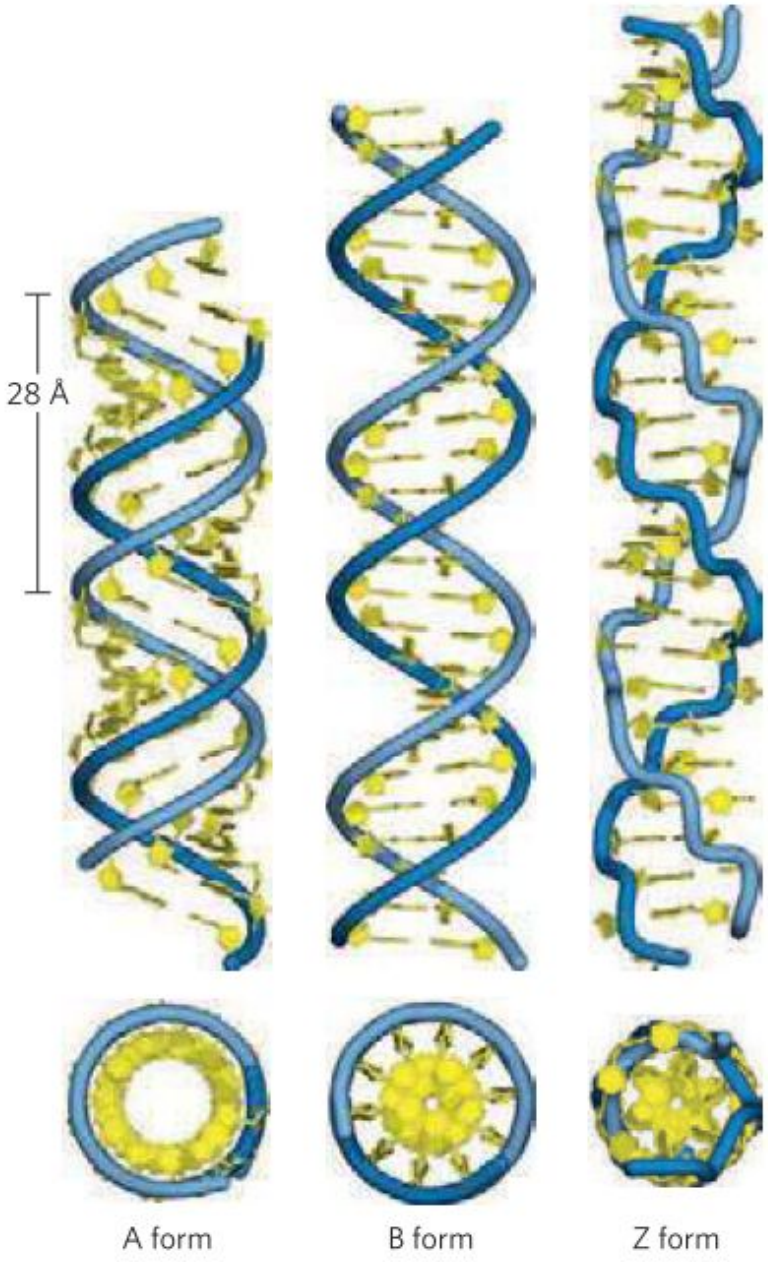


FIGURE 8–17 Comparison of A, B, and Z forms of DNA. Each structure shown here has 36 base pairs. The riboses and bases are shown in yellow. The phosphodiester backbone is represented as a blue rope. Blue is the color used to represent DNA strands in later chapters. The table summarizes some properties of the three forms of DNA.

The Watson and Crick's structure is also referred to as the B form and it is the most stable structure under physiological conditions.

A form is preferred in many solutions which are devoid of water.

	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3′ endo	C-2′ endo	C-2′ endo for pyrimidines; C-3′ endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrimidines; syn for purines

